SHARP SIEVE INEQUALITY IN FINITE FIELD PROJECTION THEORY WITH SPACING CONDITIONS

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Abstract. We prove a projection estimate in finite fields $\mathbb{F}_p$ for p prime by using ideas of Linnik's large sieve. The result we obtain gives a sharp estimate for cases when the set X is concentrated. Hence, our bound is sharp for the subgrid example.

1. Introduction

Let's formulate projection theory in $\mathbb{F}_p$ with p prime with the identification $\mathbb{F}_p \sim \mathbb{Z}/p\mathbb{Z}$: we have $X \subset (\mathbb{Z}/p\mathbb{Z})^2$ a set of points and $D \subset \mathbb{Z}/p\mathbb{Z}$ a set of directions. For each direction $\theta \in D$, we define $\pi_\theta : (\mathbb{Z}/p\mathbb{Z})^2 \to \mathbb{Z}/p\mathbb{Z}$ as $\pi_\theta(x) = x_1 + \theta x_2$. Then the size of the biggest projection is $S = S(X, D) = \max_{\theta \in D} |\pi_\theta(X)|$.

In this paper, we aim to make progress towards the following conjecture:

Conjecture 1.1. *For p prime, we have $|D| \lesssim \frac{S^2}{|X|}$.*

Our approach will be to use ideas of sieve theory, such as that of the Linnik large sieve, to get a sieve type inequality in the finite field setting. We first recall Linnik's theorem:

Theorem 1.2 (Linnik's large sieve)**.** *Let $f : \mathbb{Z} \to \mathbb{C}$ be a function supported in $[0, N]$. And let $P_M = \{p \in \mathbb{P} : \frac{M}{2} < p \leq M\}$. Then for any M:*

$$\sum_{p \in P_M} \|(\pi_p f)_H\|^2_{L^2(\mathbb{Z}/p\mathbb{Z})} \lesssim \max\Big(M, \frac{N}{M}\Big)\|f\|^2_{L^2(\mathbb{Z})} \tag{1.1}$$

The above statement uses the following definitions:

Definition 1.3. Let $f : \mathbb{Z} \to \mathbb{C}$ and p be prime. Then $\pi_p f : \mathbb{Z}/p\mathbb{Z} \to \mathbb{C}$ is defined as

$$\pi_p f(x) = \sum_{y \equiv x \bmod p} f(y) \tag{1.2}$$

Definition 1.4. Let $g : \mathbb{Z}/p\mathbb{Z} \to \mathbb{C}$ be a function. We let g_0 and g_h be functions from $\mathbb{Z}^2$ to $\mathbb{C}$ defined as

$$g_0(x) = \frac{1}{p} \sum_{x \in \mathbb{Z}/p\mathbb{Z}} g(x) \tag{1.3}$$

and $g_h = g - g_0$. We say that g_0 is *the low frequency part* or *the constant part* of g and g_H is *the high frequency part* of g.

We now give a definition useful for our paper:

Definition 1.5. Let $\tilde{\theta}$ be an integer. We let $\tilde{\pi}_{\tilde{\theta}} : \mathbb{Z}^2 \to \mathbb{Z}$ be the projection map $\tilde{\pi}_{\tilde{\theta}}(x) = x_1 + \tilde{\theta} x_2$. For any function $f : \mathbb{Z}^2 \to \mathbb{C}$ we define the projection $\tilde{\pi}_{\tilde{\theta}} f : \mathbb{Z} \to \mathbb{C}$ as

$$\tilde{\pi}_{\tilde{\theta}} f(x) = \sum_{\tilde{\pi}_{\tilde{\theta}}(y)=x} f(y) \tag{1.4}$$

Let us explain how the above definition is useful: recall our projection theory setup with $X \subset (\mathbb{Z}/p\mathbb{Z})^2$, $D \subset \mathbb{Z}/p\mathbb{Z}$ and $S = S(X, D)$. For all $x \in \mathbb{Z}/p\mathbb{Z}$, let $\tilde{x} \in [0, p-1]$ be the coset representative of x, and say $\tilde{X} = \{\tilde{x} = (\tilde{x_1}, \tilde{x_2}) : x \in X\} \subset [0, p-1]^2$. Similarly let $\tilde{D} = \{\tilde{\theta} : \theta \in D\} \subset [0, p-1]$. We can then notice from the definitions that $\pi_p \tilde{\pi}_{\tilde{\theta}} \mathbf{1}_{\tilde{X}} = \pi_\theta \mathbf{1}_X$. Furthermore, for all $x \in \mathbb{Z}/p\mathbb{Z}$:

$$\begin{aligned} \pi_p \tilde{\pi}_{\tilde{\theta}} \mathbf{1}_{\tilde{X}}(x) &= \pi_\theta \mathbf{1}_X(x) \\ &= |\{y \in (\mathbb{Z}/p\mathbb{Z})^2 : \pi_\theta(y) = x\}| \text{ has support of size } \leq S \end{aligned} \tag{1.5}$$

It turns out that a proper way to exploit the fact that a function on $\mathbb{Z}/p\mathbb{Z}$ has a small support is to divide it into the *high frequency* and *low frequency* parts:

Lemma 1.6. *Let $g : \mathbb{Z}/p\mathbb{Z} \to \mathbb{C}$ be a non-negative function with $|\mathrm{supp}\, g| \leq S$ and $\|g\|_{L^1} = M$. If $S \leq \frac{1}{2}p$ then $\|g_h\|_{L^2}^2 \gtrsim \frac{M^2}{S}$.*

In conclusion, we will show that our projection theory setup will give that whenever $S \leq \frac{1}{2}p$:

$$\sum_{\theta \in D} \|(\pi_p \tilde{\pi}_{\tilde{\theta}} \mathbf{1}_{\tilde{X}})_H\|_{L^2}^2 \gtrsim |D| \frac{|X|^2}{S} \tag{1.6}$$

Now we are ready to state the main technical result of this paper, which is our sieve inequality:

Theorem 1.7 (Main Sieve inequality)**.** *Let $f : \mathbb{Z}^2 \to \mathbb{C}$ be a function supported on a set X, where $X \subset [0, p-1]^2$ and p an odd prime. Let $D \subset [0, p-1]$ be a set of directions. Let $T = T(X)$ be the smallest positive integer such that the set X can be contained in a square $Q = Q(X)$ of side lengths $T \times T$. We have*

$$\sum_{\theta \in D} \|(\pi_p \tilde{\pi}_\theta f)_H\|_{L^2(\mathbb{Z}/p\mathbb{Z})}^2 \lesssim T |D|^{\frac{1}{2}} \|f\|_{L^2(\mathbb{Z}^2)}^2 \tag{1.7}$$

By comparing the above Theorem 1.7 with the bound (1.6) we obtain the following projection estimate:

Theorem 1.8. *Let p be a prime and $X \subset \mathbb{F}_p^2$, $D \subset \mathbb{F}_p$ with $S = S(X, D)$. Let $T = T(X)$ be the smallest positive integer such that there exists a sub-square in $\mathbb{F}_p^2$ of side lengths $T \times T$ containing X. Then*

$$|D| \lesssim \frac{S^2 T^2}{|X|^2} \tag{1.8}$$

Remark 1.9. Even though this bound is weaker than the desired bound $|D| \lesssim \frac{S^2}{|X|}$ since $|X| \leq T^2$, it is still **sharp** in our subgrid example. In fact, for a subgrid $X = \{(x_1, x_2) : 1 \leq x_1, x_2 \leq N\}$ and directions $D = \{\frac{a_1}{a_2} : 1 \leq a_1, a_2 \leq A\}$, we know

$S \sim AN$ and clearly $T = N$. So $T^2 = |X|$ and thus $\frac{S^2T^2}{|X|^2} = \frac{A^2N^2 \cdot N^2}{(N^2)^2} = A^2 = |D|$. Hence, Theorem 1.8 will imply our desired bound in Conjecture 1.1 if $|X| \sim T^2$, i.e. if X has some positive constant density in Q.

For the rest of the paper, we mainly focus on proving Theorem 1.7 and, after that, we finish up the details of Theorem 1.8.

2. Dictionary lemmas

In this section, we introduce the so-called *Dictionary* lemmas that give relations between Fourier transforms of functions and its projections.

Lemma 2.1. *Let $f : \mathbb{Z}^2 \to \mathbb{C}$ be a function, θ be an integer and let $\pi_\theta f : \mathbb{Z} \to \mathbb{C}$ be its projection. Then for $\xi \in \mathbb{T}$ we have*

$$\widehat{\pi_\theta f}(\xi) = \widehat{f}(\xi_\theta) \text{ where } \xi_\theta = (\xi, \theta\xi) \in \mathbb{T}^2 \tag{2.1}$$

Proof. We calculate that

$$\begin{aligned} \widehat{\pi_\theta f}(\xi) &= \sum_{x\in\mathbb{Z}} \pi_\theta f(x) e^{-2\pi i x\xi} \\ &= \sum_{x\in\mathbb{Z}} \left(\sum_{\pi_\theta(y)=x} f(y) \right) e^{-2\pi i x\xi} \\ &= \sum_{y\in\mathbb{Z}^2} f(y) e^{-2\pi i(\pi_\theta(y)\cdot\xi)} \\ &= \sum_{y\in\mathbb{Z}^2} f(y) e^{-2\pi i y\cdot\xi_\theta} = \widehat{f}(\xi_\theta) \end{aligned} \tag{2.2}$$

where we used the observation that $\pi_\theta(y)\cdot\xi = (y_1 + \theta y_2)\cdot\xi = y_1\xi + \theta y_2\xi = y \cdot \xi_\theta$. □

Lemma 2.2. *Let $g : \mathbb{Z} \to \mathbb{C}$ be a function, p a prime and $\pi_p g : \mathbb{Z}/p\mathbb{Z} \to \mathbb{C}$ be its projection. Then for $\alpha \in \mathbb{Z}/p\mathbb{Z}$ we have*

$$\widehat{\pi_p g}(\xi) = \widehat{g}\left(\frac{\xi}{p}\right) = \widehat{g}\left(\frac{\tilde{\xi}}{p}\right) \tag{2.3}$$

Proof. Similarly, we find that

$$\begin{aligned} \widehat{\pi_p g}(\xi) &= \sum_{a\in\mathbb{Z}/p\mathbb{Z}} \pi_p g(a) e^{-2\pi i \frac{a\xi}{p}} \\ &= \sum_{a\in\mathbb{Z}/p\mathbb{Z}} \left(\sum_{x\equiv a \bmod p} g(x) \right) e^{-2\pi i \frac{a\xi}{p}} \\ &= \sum_{x\in\mathbb{Z}} g(x) e^{-2\pi i \frac{x\xi}{p}} = \widehat{g}\left(\frac{\xi}{p}\right) \end{aligned} \tag{2.4}$$

(2.5)

as desired. □

Lemma 2.3. *Let $g : \mathbb{Z} \to \mathbb{C}$ be a function, and we write $g = g_0 + g_h$ as defined in Definition 1.4. Then*

$$\widehat{g_h}(\alpha) = \begin{cases} 0 & \text{if } \alpha = 0 \\ \widehat{g}(\alpha) & \text{if } \alpha \neq 0 \end{cases} \tag{2.6}$$

Proof. We calculate that

$$\widehat{g_0}(0) = \sum_{x\in\mathbb{Z}/p\mathbb{Z}} g_0(x) = p\cdot\frac{1}{p}\sum_{x\in\mathbb{Z}/p\mathbb{Z}} g(x) = \widehat{g}(0) \tag{2.7}$$

and thus $\widehat{g_h}(0) = \widehat{g}(0) - \widehat{g_0}(0) = 0$. Furthermore, for $\alpha \neq 0$ and $\alpha \in \mathbb{Z}/p\mathbb{Z}$, we get

$$\widehat{g_0}(\alpha) = \sum_{x\in\mathbb{Z}/p\mathbb{Z}} g_0(x)e^{-2\pi i\frac{x\alpha}{p}} = g_0\sum_{x\in\mathbb{Z}/p\mathbb{Z}} e^{-2\pi i\frac{x\alpha}{p}} = g_0\frac{1-e^{2\pi i\frac{p\alpha}{p}}}{1-e^{-2\pi i\frac{\alpha}{p}}} = 0 \tag{2.8}$$

and so $\widehat{g_h}(\alpha) = \widehat{g}(\alpha) - \widehat{g_0}(\alpha) = \widehat{g}(\alpha)$. □

Corollary 2.4. *Putting those together, we find that for any $f : \mathbb{Z}^2 \to \mathbb{C}$ function, the identity*

$$\widehat{(\pi_p\tilde{\pi}_\theta f)}_H(\alpha) = \widehat{f}\Big(\frac{\alpha}{p}, \frac{\alpha\theta}{p}\Big) \tag{2.9}$$

holds for all $\alpha \neq 0$ and $\widehat{(\pi_p\tilde{\pi}_\theta f)}_H(\alpha) = 0$ for $\alpha = 0$. Thus by Plancherel we get

$$\|(\pi_p\tilde{\pi}_\theta f)_H\|^2_{L^2(\mathbb{Z}/p\mathbb{Z})} = \frac{1}{p}\sum_{\alpha=1}^{p-1}\Big|\widehat{f}\Big(\frac{\alpha}{p}, \frac{\alpha\theta}{p}\Big)\Big|^2 \tag{2.10}$$

Finally, this implies that

$$\begin{aligned}\sum_{\theta\in D}\|(\pi_p\tilde{\pi}_\theta f)_H\|^2_{L^2(\mathbb{Z}/p\mathbb{Z})} &= \frac{1}{p}\sum_{\theta\in D}\sum_{\alpha=1}^{p-1}\Big|\widehat{f}\Big(\frac{\alpha}{p}, \frac{\alpha\theta}{p}\Big)\Big|^2 \\ &= \frac{1}{p}\sum_{\xi\in Q_D}\big|\widehat{f}(\xi)\big|^2\end{aligned} \tag{2.11}$$

for the finite set $Q_D \subset \mathbb{T}^2$ defined as $Q_D = \{(\frac{\alpha}{p}, \frac{\alpha\theta}{p}) : \theta \in D, 1 \leq \alpha \leq p-1\}$

3. Locally constant property

In light of Corollary 2.4, to prove Theorem 1.7 we want to

$$\text{compare } \frac{1}{p}\sum_{\xi\in Q_D}\big|\widehat{f}(\xi)\big|^2 \text{ with } \int_{\mathbb{T}^2}\big|\widehat{f}(\omega)\big|^2 \mathrm{d}\omega = \|f\|^2_{L^2(\mathbb{Z}^2)} \tag{3.1}$$

This reminds us of Riemann integration. How we accomplish the above task (3.1) will be to utilize what we call the *locally constant* property for $\widehat{f}$.

Theorem 3.1. *For the $T \times T$ square $Q = Q(X)$ containing X, there exists a function $\eta : \mathbb{Z}^2 \to \mathbb{C}$ with the following properties:*

(i) *$\eta = 1$ on the $T \times T$ square Q and $|\eta|$ rapidly decays off outside of Q.*
(ii) *$|\widehat{\eta}| \sim T^2$ on $[-\frac{1}{T}, \frac{1}{T}]^2$.*
(iii) *$\widehat{\eta}$ is essentially supported in the region $[-\frac{1}{T}, \frac{1}{T}]^2$ in the following sense: if $\xi \notin [-\frac{1}{T}, \frac{1}{T}]^2$ we have*

$$\big|\widehat{\eta}(\xi)\big| \lesssim T^2(T|\xi|_*)^{-1000} \tag{3.2}$$

where we denoted $|\xi|_* = \max(|\xi_1|, |\xi_2|)$ with $\xi_1, \xi_2 \in [-\frac{1}{2}, \frac{1}{2}]$.

Proof. **TO DO.** □

We want to explain how the function η helps with (3.1). Notice that since $\eta = 1$ on X and the function $f : \mathbb{Z}^2 \to \mathbb{C}$ we consider in Theorem 1.7 is supported in X, we have $f = f\eta$. Thus by taking the Fourier Transform and applying triangle inequality gives

$$|\widehat{f}(\xi)| \leq \int_{\mathbb{T}^2} |\widehat{f}(\omega)||\widehat{\eta}(\xi-\omega)|\mathrm{d}\omega \tag{3.3}$$

This is the so-called *locally constant* property for $\widehat{f}$: since $\widehat{\eta}$ is essentially supported in $[-\frac{1}{T}, \frac{1}{T}]^2$ with value $\sim T^2$ on this set, we get that (3.3) is saying that $|\widehat{f}(\xi)|$ is roughly equal to the average of $|\widehat{f}|$ on a $[-\frac{1}{T}, \frac{1}{T}]^2$ square around ξ. So heuristically we can think that $\widehat{f}$ is locally constant on small squares of size $\frac{1}{T} \times \frac{1}{T}$ inside $\mathbb{T}^2$.

Next, we want to obtain a similar inequality to (3.3) for $|\widehat{f}|^2$ instead. Square both sides of (3.3) and use Cauchy-Schwartz to get

$$|\widehat{f}(\xi)|^2 \leq \left(\int_{\mathbb{T}^2} |\widehat{f}(\omega)|^2 |\eta(\xi-\omega)|\mathrm{d}\omega\right)\left(\int_{\mathbb{T}^2} |\widehat{\eta}(\xi-\omega)|\mathrm{d}\omega\right) \tag{3.4}$$

Proposition 3.2. *We have* $\int_{\mathbb{T}^2} |\widehat{\eta}(\omega)|\mathrm{d}\omega \lesssim 1$.

Proof. Heuristically speaking, $\widehat{\eta}$ is a step function of value T^2 on the square $[-\frac{1}{T}, \frac{1}{T}]^2$. Hence we find that $|\widehat{\eta}|$ integrates to $\lesssim 1$ on this region. Since $|\widehat{\eta}|$ decays outside the region $[-\frac{1}{T}, \frac{1}{T}]^2$ we expect that the contribution from that region is very small.

To make this precise, we define the regions

$$R_k = \{\xi \in \mathbb{T}^2 : \frac{1}{T}k \leq |\xi|_* \leq \frac{1}{T}(k+1)\} \subset \mathbb{T}^2 \tag{3.5}$$

for $k \geq 0$. Clearly $R_0 = [-\frac{1}{T}, \frac{1}{T}]^2$ and we have seen that

$$\int_{R_0} |\widehat{\eta}(\omega)|\mathrm{d}\omega \lesssim T^2 \cdot |R_0| = 1 \tag{3.6}$$

Now notice for $\omega \in R_k$ for $k \geq 1$ we have $|\widehat{\eta}(\omega)| \lesssim T^2(T|\omega|_*)^{-1000} \leq T^2(k+1)^{-1000}$. Hence we get

$$\int_{R_k} |\widehat{\eta}(\omega)|\mathrm{d}\omega \lesssim T^2(k+1)^{-1000}|R_k| \tag{3.7}$$

$$\leq T^2(k+1)^{-1000}\left(\frac{1}{T}(k+1)\right)^2 = (k+1)^{-998} \tag{3.8}$$

Hence it's clear that

$$\sum_{k\geq 1}\int_{R_k} |\widehat{\eta}(\omega)|\mathrm{d}\omega \lesssim \sum_{k\geq 1}\frac{1}{(k+1)^{998}} \lesssim 1 \tag{3.9}$$

Putting those together we get $\int_{\mathbb{T}^2} |\widehat{\eta}| \lesssim 1$ as desired. □

After this Proposition we see that (3.4) becomes

$$|\widehat{f}(\xi)|^2 \lesssim \int_{\mathbb{T}^2} |\widehat{f}(\omega)|^2 |\widehat{\eta}(\xi-\omega)|\mathrm{d}\omega \tag{3.10}$$

This is the locally constant property we want for $\left|\widehat{f}\right|^2$. Using this, we can bound our desired quantity in (3.1) as

$$\begin{aligned}\frac{1}{p}\sum_{\xi\in Q_D}\left|\widehat{f}(\xi)\right|^2 &\lesssim \frac{1}{p}\sum_{\xi\in Q_D}\int_{\mathbb{T}^2}\left|\widehat{f}(\omega)\right|^2\left|\widehat{\eta}(\xi-\omega)\right|\mathrm{d}\omega \\ &= \frac{1}{p}\int_{\mathbb{T}^2}\left|\widehat{f}(\omega)\right|^2\left(\sum_{\xi\in Q_D}\left|\widehat{\eta}(\xi-\omega)\right|\right)\mathrm{d}\omega\end{aligned} \tag{3.11}$$

4. Geometric counting lemma

After (3.11), we want to obtain a uniform bound on $\sum_{\xi\in Q_D}\left|\widehat{\eta}(\xi-\omega)\right|$ over all $\omega\in\mathbb{T}^2$. Since $\left|\widehat{\eta}\right|$ is essentially supported in $[-\frac{1}{T},\frac{1}{T}]^2\subset\mathbb{T}^2$ we see that we will need to count the number of $\xi\in Q_D$ with $\xi-\omega\in[-\frac{1}{T},\frac{1}{T}]^2$ for each fixed ω. This motivates our *geometric counting lemma* 4.3.

First we give the following definition:

Definition 4.1. Let $L_{\theta,p}$ be the lattice consisting of points $x\in\mathbb{Z}^2$ with $x_2\equiv\theta x_1 \bmod p$. For any point $x:\mathbb{Z}^2$ we write $|x|_*=\max(|x_1|,|x_2|)$. Let

$$r_{\theta,p}=\min_{x\in L_{\theta,p}\setminus\{0\}}|x|_* \tag{4.1}$$

Remark 4.2. It's not hard to see that $L_{\theta,p}$ is in fact a lattice: for two points $x,y\in L_{\theta,p}$ we have that $x_2+y_2\equiv\theta x_1+\theta y_2 \bmod p$.

Lemma 4.3 (Geometric counting lemma)**.** *For any square region A inside $\mathbb{T}^2$ of size $R\times R$ we have*

$$|Q_D\cap A|\lesssim\sum_{\theta\in D}\lceil Rr_\theta\rceil\cdot\left\lceil\frac{pR}{r_\theta}\right\rceil \tag{4.2}$$

Proof. We first give a characterization of the points Q_D. Let's consider the set

$$Q_\theta=\{\xi\in Q_D:\xi_2=\theta\xi_1\}=\left\{\left(\frac{\alpha}{p},\frac{\alpha\theta}{p}\right):1\le\alpha\le p-1\right\}\subset\mathbb{T}^2 \tag{4.3}$$

We will prove the following, which suffices for (4.2) since $Q_D=\cup_{\theta\in D}Q$:

$$|Q_\theta\cap A|\lesssim\lceil Rr_\theta\rceil\cdot\left\lceil\frac{R}{r_\theta}\right\rceil \tag{4.4}$$

Notice that for any point $(\frac{\alpha}{p},\frac{\alpha\theta}{p})$ in $Q_\theta\subset\mathbb{T}^2$, the left $\mathbb{Z}^2$-coset it represents is

$$\left(\frac{\alpha}{p},\frac{\alpha\theta}{p}\right)+\mathbb{Z}^2=\left\{\left(\frac{\alpha}{p}+z_1,\frac{\alpha\theta}{p}+z_2\right):z_1,z_2\in\mathbb{Z}\right\} \tag{4.5}$$

$$=\left\{\left(\frac{x_1}{p},\frac{x_2}{p}\right):x_2\equiv\theta x_1 \bmod p\right\}=\frac{1}{p}L_{\theta,p} \tag{4.6}$$

TO FINISH. □

5. Proof of Main technical result

In this section, we will give the proof of Theorem 1.7 using all of the tools we've built so far.

Recalling (3.11), we claim that showing the following Proposition will be enough for the proof of Theorem 1.7.

Proposition 5.1. *We have*

$$\sum_{\xi \in Q_D} \left|\widehat{\eta}(\xi-\omega)\right| \lesssim pT|D|^{\frac{1}{2}} \tag{5.1}$$

for every $\omega \in \mathbb{T}^2$.

Indeed, we can see that Proposition 5.1 and (**??**) together implies

$$\begin{aligned} \sum_{\theta \in D} \|(\pi_p \tilde{\pi}_\theta f)_H\|^2_{L^2(\mathbb{Z}/p\mathbb{Z})} &= \frac{1}{p} \sum_{\xi \in Q_D} \left|\widehat{f}(\xi)\right|^2 \\ &\lesssim \frac{1}{p} \int_{\mathbb{T}^2} \left|\widehat{f}(\omega)\right|^2 \cdot pT|D|^{\frac{1}{2}} \mathrm{d}\omega \\ &= T|D|^{\frac{1}{2}} \|f\|^2_{L^2(\mathbb{Z}^2)} \end{aligned} \tag{5.2}$$

by Plancherel, proving our main technical result Theorem 1.7.

Proof of Proposition 4.2. We utilize the division of $\mathbb{T}^2$ into regions R_k for $k \geq 0$ that we defined in the proof of Proposition 3.2. First, let's focus on the region $R_0 = [-\frac{1}{T}, \frac{1}{T}]^2$:

$$\sum_{\xi \in Q_D, \xi-\omega \in R_0} || \tag{5.3}$$

$\square$

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